

Algorithm

Step 1: Start

Begin the program.

Step 2: Load Required Packages

Load the required R packages:

- data.table
- ggplot2

Step 3: Create Project Directories

Check whether the Output folder exists.

- If it does not exist, create it.
- Set the working directory to the project folder.

Step 4: Generate Sales Dataset

Generate **5,000 sales transaction records** with:

- Transaction ID
- Customer ID
- Product ID
- City
- Category
- Quantity
- Price
- Payment Method
- Date

Calculate:

Total Amount = Quantity × Price

Store the generated data as:

sales_transactions.csv

Step 5: Import Dataset

Import the CSV file using:

fread()

Store the data in a data.table object.

Step 6: Explore the Dataset

Perform initial data exploration:

1. Display the first few records.
2. Check the structure of the dataset.
3. Generate summary statistics.
4. Check for missing values.
5. Detect duplicate records.

Step 7: Clean the Dataset

Clean the data by:

1. Removing duplicate rows.
2. Removing rows containing missing values.
3. Validating the cleaned dataset.

Step 8: Perform Data Transformation

Create new columns using the := operator.

Calculate:

$\text{Discount} = \text{Total Amount} \times \text{Discount Rate}$

$\text{Net Amount} = \text{Total Amount} - \text{Discount}$

Extract the month from the date:

$\text{Month} = \text{Month}(\text{Date})$

Step 9: Apply In-Place Updates

Use the := operator to update or create columns directly inside the data.table.

This avoids unnecessary copying of the dataset and improves performance.

Step 10: Create an Index

Use:

setkey()

to create an index on an appropriate column such as:

Customer_ID

This improves data searching and query performance.

Step 11: Generate Summary Reports

Perform grouping and aggregation.

Customer Summary

Group by Customer_ID and calculate:

- Total Transactions
- Total Quantity
- Total Revenue
- Total Net Amount

Product Summary

Group by Product_ID and calculate:

- Total Quantity Sold
- Total Revenue
- Average Price

City Summary

Group by City and calculate:

- Number of Transactions
- Total Quantity
- Total Revenue

Category Summary

Group by Category and calculate:

- Number of Transactions
- Total Quantity
- Total Revenue

Step 12: Generate Business Insights

Calculate:

Total Revenue

Total Number of Transactions

Find:

- Best-performing city
- Best-performing category
- Top customer
- Most-used payment method

Step 13: Generate Visualizations

Create the following visualizations using ggplot2:

1. City Transactions

- Display the number of transactions in each city.

2. Category Revenue

- Display revenue generated by each category.

3. Top Customers

- Display the customers with the highest revenue.

4. Payment Distribution

- Display the distribution of different payment methods.

Step 14: Export Results

Save the summary reports as CSV files:

customer_summary.csv

product_summary.csv

city_summary.csv

category_summary.csv

Save the charts as PNG files inside the Output folder.

Step 15: End

End the program.

Simple Algorithm Flow

START

↓

Load data.table and ggplot2

↓

Create 5,000 sales transactions

↓

Save dataset as CSV

↓

Import dataset using fread()

↓

Explore dataset

↓

Check missing values and duplicates

↓

Clean the dataset

↓

Create Discount, Net_Amount, and Month columns

↓

Use := for in-place updates

↓

Create index using setkey()

↓

Group and aggregate data

↓

Generate business summaries

↓

Create visualizations

↓

Export CSV reports and PNG charts

↓

END